How Much of Reported Quantization Degradation Is Measurement? Decomposing Accuracy Loss into Competence, Compliance, and Truncation

Anonymous ACL submission

Abstract

[**TODO:** Abstract — written in W5 (C7), after results freeze.]

1 Introduction

2 Related Work

A live contradiction about generation length. Whether low-bit quantization lengthens chain-of-thought output is currently disputed. Lian et al. (2026) report that INT4/INT3 quantization can preserve accuracy while *increasing* reasoning-token usage, and propose a CoT Token Inflation Ratio to quantify it; Lotfi et al. (2026) likewise find that aggressive post-training quantization lengthens CoT. Liu et al. (2025) report the opposite: "contrary to expectations, quantized models do not exhibit increased output lengths." We take no position on which generator behaviour is correct, because the disagreement has a consequence that neither side measures. Under a *fixed* generation cap, a model that emits longer traces is scored wrong more often whenever its answer falls beyond the cap — a scoring outcome, not a competence one. No study on either side of this dispute sweeps the generation cap, so no study can separate the two. That separation is our H1.

Degradation has already been partly explained — by other causes. Kurtic et al. (2025) show that reported quantization degradation was overstated, and attribute the excess to untuned quantization-side hyperparameters. We treat that result as established and *prior*, not as a target: our claim concerns the residual that remains after quantization-side tuning, and our mechanism is orthogonal to theirs. Lotfi et al. (2026) supply the most direct antecedent for the present work, finding that in as many as 52% of quantized-model failures the model reaches the correct answer in an intermediate step but does not emit it as the final answer. Their interest is in fixing the generator, via a logit penalty on overthinking markers; ours is in what such failures do to a reported accuracy delta when they are left in place.

The centre of gravity runs the other way. The dominant finding in this literature is that benchmarks *understate* quantization damage, not overstate it. Marchisio et al. (2024) find a 1.7% automatic drop in Japanese corresponds to 16.0% by human judgement, and Dutta et al. (2024) show that models matched on accuracy still diverge behaviourally — the "flips" framing, which is theirs and which we do not claim. We regard both directions as simultaneously true: an accuracy delta can over-penalise formatting *and* under-detect behavioural change, because these are different quantities. Our contribution is to the first, and it does not license any conclusion about the second.

Calibrating extraction, not loosening it. An argument that evaluation should simply be more permissive is already refuted. Ren et al. (2026) find that 8.2–44.1% of answers graded correct are reached by shortcut hacking, so lenient extraction over-credits as surely as strict extraction under-credits. We therefore treat the strict/permissive contrast as a measurement instrument to be characterised rather than a knob to be turned in one direction, and we report both rungs throughout.

Prompt- and protocol-sensitivity. Hua et al. (2025) ask whether prompt sensitivity is a genuine model property or an artifact of evaluation, and we are structurally the sequel: they vary the prompt, we vary the scorer, and both ask whether a reported effect survives the instrument. Their null result on MATH deserves emphasis rather than a footnote — extraction there already uses symbolic equivalence, so a well-engineered verifier leaves little for a protocol effect to explain.

This bounds our claim rather than supporting it, and it is why our task selection targets benchmarks whose canonical extractors are regular expressions. Sclar et al. (2024) establish that spurious formatting choices move scores substantially, and Jo et al. (2025) and Yu et al. (2025) show that answer extraction from reasoning traces is itself an unreliable step. Schaeffer et al. (2023) provide the methodological template we follow: a headline effect reattributed, in part, to the measurement instrument. The lm-evaluation-harness (Biderman et al., 2024) is our instrument, and its canonical filters are the object of study rather than an implementation detail.

Serving configuration is a confound we pin, not an axis we explore. Pape et al. (2026) find that inference-backend choice alone moves benchmark scores by up to 16.6 points, and Masoudian et al. (2026) attribute roughly 39% of out-of-the-box variance to it. We read this as a reason to freeze the engine, version, and every decoding parameter across all arms, which we do and report; a floating serving configuration would make precision effects unidentifiable. We claim no novelty on this axis.

Adjacent evaluations. Broad compression benchmarks (von Rad et al., 2026; Wang et al., 2025) and task-difficulty studies (Lee et al., 2025) establish the accuracy landscape we work inside; Lee et al. (2025) in particular report that quantization disrupts instruction-following, which is the strongest available objection to our framing and which our compliance/competence split is designed to address directly rather than deflect. Prasad and Pal (2026) run a factorial quantization analysis, so factorial design is not our contribution; their outcome domain is safety alignment under sampling temperature, which is disjoint from ours. Kurt (2026) report the observation this paper sets out to explain — strict-versus-flexible GSM8K gaps of 41 to 64 points across fourteen `llama.cpp` quantizations — without interpreting the gap; we reproduce their numbers as an external anchor. Oladri et al. (2026) and Rasmussen and Qin (2026) report related failure-mode and scoring-rule effects, the former over FP32/FP16/NF4 only and so without bearing on GPTQ or AWQ.

What is new here. Permissive extraction is well known to raise absolute scores. What has not been established is whether it changes the quantization *difference*: an artifact in a level need not be an artifact in a delta, and the two are routinely conflated. We decompose the measured strict delta into competence, compliance, and truncation components, pre-register the decomposition before any generation exists, and report the residual precision effect that survives it.

3 Method

3.1 Experimental design

3.2 Quantization and calibration

3.3 The scoring ladder

3.4 Compliance, competence, and truncation

3.5 Serving configuration and determinism

4 Results

4.1 Truncation (H1)

[**TODO:** H1 — degradation vs generation budget; mean/median n_gen_tokens by precision with CIs, settling 2606.25519 vs 2504.04823.]

4.2 Recoverable failures (H2)

[**TODO:** H2 — gold-in-trace rate, permissive and conservative variants, always reported net of the incidental-match false-positive floor.]

4.3 Precision × scoring interaction (H3)

[**TODO:** H3 — GLMM interaction as an odds ratio and as a marginal effect in accuracy points with 95% CI, plus paired McNemar. A NULL IS PUBLISHABLE and must not read as failure.]

4.4 Decomposition (H4)

[**TODO:** H4 — THE CENTREPIECE TABLE. Design it first and let it drive what gets logged. Always-compliant stratum size as a first-class number; compliance-transition matrix; Manski bounds.]

4.5 Prompt format (H5) and the structural control (H6)

[**TODO:** H5 zero-shot arm; H6 ARC-Challenge under log-likelihood vs generative scoring on the same 500 items — the gap between modes IS the extraction cost.]

4.6 Statistical resolution of published deltas (H7)

[**TODO:** H7 — scrupulously neutral: "many papers report aggregate deltas without a paired test", NEVER "the deltas are noise".]

4.7 Bitwise reproducibility of quantized serving (H8)

[**TODO:** H8 — per-format bitwise disagreement under a batch-invariant engine. Self-contained result nobody has reported; may become a headline.]

5 Discussion

6 Conclusion

Limitations

Ethics Statement

Acknowledgments

References

Stella Biderman, Hailey Schoelkopf, Lintang Sutawika, Leo Gao, Jonathan Tow, Baber Abbasi, Alham Fikri Aji, Pawan Sasanka Ammanamanchi, Sidney Black, Jordan Clive, Anthony DiPofi, Julen Etxaniz, Benjamin Fattori, Jessica Zosa Forde, Charles Foster, Jeffrey Hsu, Mimansa Jaiswal, Wilson Y. Lee, Haonan Li, and 11 others. 2024. Lessons from the trenches on reproducible evaluation of language models. *arXiv preprint arXiv:2405.14782*.

Abhinav Dutta, Sanjeev Krishnan, Nipun Kwatra, and Ramachandran Ramjee. 2024. Accuracy is not all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Andong Hua, Kenan Tang, Chenhe Gu, Jindong Gu, Eric Wong, and Yao Qin. 2025. Flaw or artifact? rethinking prompt sensitivity in evaluating llms. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Hwiyeol Jo, Joosung Lee, Jaehone Lee, Sang-Woo Lee, Joonsuk Park, and Kang Min Yoo. 2025. Finding answers in thought matters: Revisiting evaluation on large language models with reasoning. *arXiv preprint arXiv:2510.14773*.

Uygar Kurt. 2026. Which quantization should i use? a unified evaluation of llama.cpp quantization on llama-3.1-8b-instruct. *arXiv preprint arXiv:2601.14277*.

Eldar Kurtic, Alexandre Marques, Shubhra Pandit, Mark Kurtz, and Dan Alistarh. 2025. "give me bf16 or give me death"? accuracy-performance trade-offs in llm quantization. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*.

Jemin Lee, Sihyeong Park, Jinse Kwon, Jihun Oh, and Yongin Kwon. 2025. Exploring the trade-offs: Quantization methods, task difficulty, and model size in large language models from edge to giant. In *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence (IJCAI)*.

Xinyu Lian, Walid Krichene, Beichen Huang, Masahiro Tanaka, Olatunji Ruwase, Li Zhang, and Minjia Zhang. 2026. Quantization inflates reasoning: Token inflation as a hidden cost of low-bit reasoning models. *arXiv preprint arXiv:2606.25519*.

Ruikang Liu, Yuxuan Sun, Manyi Zhang, Haoli Bai, Xianzhi Yu, Tiezheng Yu, Chun Yuan, and Lu Hou. 2025. Quantization hurts reasoning? an empirical study on quantized reasoning models. In *Conference on Language Modeling (COLM)*.

Sanae Lotfi, Polina Kirichenko, Steven Li, and Zechun Liu. 2026. Quantized reasoning models think they need to think longer, but they do not. *arXiv preprint arXiv:2606.00206*.

Kelly Marchisio, Saurabh Dash, Hongyu Chen, Dennis Aumiller, Ahmet Üstün, Sara Hooker, and Sebastian Ruder. 2024. How does quantization affect multilingual llms? In *Findings of the Association for Computational Linguistics: EMNLP 2024*.

Shahed Masoudian, Passant Shafaei, Monorama Swain, and Markus Schedl. 2026. What we observe as llm behavior can be a side-effect of inference backend. *arXiv preprint arXiv:2608.04714*.

Renuka Oladri, Mohan Vamsi Varadaraju Priya, and Jerry Wu. 2026. Silent failures in quantized llm reasoning: A taxonomy-based analysis of hollow convergence and failure mode shifts. *arXiv preprint arXiv:2607.09999*.

David Pape, Jonathan Evertz, and Lea Schönherr. 2026. The silent hyperparameter: Quantifying the impact of inference backends on llm reproducibility. *arXiv preprint arXiv:2605.19537*.

Hari Prasad and Ritam Pal. 2026. The joint effect of quantization and sampling temperature on llm safety alignment: A factorial analysis. *arXiv preprint arXiv:2606.29581*.

Anton Rasmussen and Hong Qin. 2026. When calibration depends on the scoring rule: Quantized biomedical llm classification. *arXiv preprint arXiv:2608.03854*.

Xuan Ren, Weiqi Zhai, Tianle Pu, Yihua Zhu, Hu Wei, and Bing Zhao. 2026. Right answer, wrong method: Shortcut hacking misleads the evaluation of llm reasoning on frontier science benchmarks. *arXiv preprint arXiv:2608.02442*. Work in progress.

Rylan Schaeffer, Brando Miranda, and Sanmi Koyejo. 2023. Are emergent abilities of large language models a mirage? In *Advances in Neural Information Processing Systems (NeurIPS)*.

Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. 2024. Quantifying language models' sensitivity to spurious features in prompt design or: How i learned to start worrying about prompt formatting. In *International Conference on Learning Representations (ICLR)*.

Jonathan von Rad, Yong Cao, and Andreas Geiger. 2026. Unicomp: A unified evaluation of large language model compression via pruning, quantization and distillation. *arXiv preprint arXiv:2602.09130*.

Jiacheng Wang, Yejun Zeng, Jinyang Guo, Yuqing Ma, Aishan Liu, and Xianglong Liu. 2025. Slmquant: Benchmarking small language model quantization for practical deployment. *arXiv preprint arXiv:2511.13023*.

Qingchen Yu, Zifan Zheng, Shichao Song, Zhiyu Li, Feiyu Xiong, Bo Tang, and Ding Chen. 2025. xfinder: Large language models as automated evaluators for reliable evaluation. In *International Conference on Learning Representations (ICLR)*.

A Frozen configuration

B Annotation rubric

C Pre-registration